Part 5 - Portfolio Analytics

Summary. The MSCI USA Value Index has the advertised positive value exposure (HML loading 0.322) and negative small-cap exposure (SMB loading -0.174). A Fama–French factor replication captures 94.84% of its monthly return variation, but MSCI's additional style-selection rules do not generate statistically significant positive alpha. The replicating portfolio has a higher gross Sharpe ratio. Separately, the equal-weighted four-factor MSCI blend earns only 0.15% annualized active return against 3.39% tracking error (information ratio 0.04). Minimum volatility is the largest contributor to active risk.

5A. What is MSCI actually selling? A value-definition audit

The non-obvious distinction is that MSCI Value is not simply a portfolio of the cheapest stocks. MSCI scores value using **book value/price, 12-month forward earnings/price and dividend yield**, while separately measuring growth through **long-term and short-term forward EPS growth, current internal growth, historical long-term EPS growth and historical long-term sales-per-share growth**. Because value and growth are evaluated separately, a stock can score well on both dimensions; the style indexes are not a binary cheap-versus-expensive split. Securities can receive partial style inclusion weights.

This is economically consequential. **Book/price** emphasizes accounting assets, **forward earnings/price** incorporates analysts' expectations of profitability, and **dividend yield** captures cash distributed to shareholders. A stock with weak book value but strong expected earnings need not be treated like a conventional high-book-to-market stock. The index is therefore a *multi-signal valuation screen*, not a pure bet on balance-sheet cheapness. MSCI also recognizes that a universal growth signal is not appropriate for every industry: the cited methodology excludes historical long-term sales-per-share growth for banks and diversified financials (subject to specified sub-industry exceptions), and excludes long-term forward EPS growth in Small Cap indexes. These are **sector/universe-specific measurement choices**, not evidence of superior performance.

Exhibit 5A. What is being measured?

Due-diligence question	MSCI USA Value	Fama–French HML
What defines cheapness?	Book/price + forward earnings/price + dividend yield	High-minus-low book-to-market portfolio spread
Is growth assessed separately?	Yes; five forecast, internal and historical growth indicators	No comparable multi-indicator growth classification
Does classification vary by security type?	Yes; sector and index-size exceptions	Standard academic size/book-to-market sorts
What is the investable object?	Long-only, large/mid-cap style index	Long-short factor return spread, not a stand-alone long-only index

Investment implication. MSCI's methodology may improve *classification* and investability, but that does not establish *alpha*. The test for a manager-selection team is whether the extra signals deliver returns that cannot be explained by inexpensive market, size and value exposures. Sections 5B–5D test precisely that. **Sources:** MSCI Global Investable Market Value and Growth Index Methodology (February 2021), section section 1.2, 2.1–2.3; Fama–French factor definitions.

5B. The index's quantitative fingerprint: advertised exposure or hidden bets?

We regress MSCI USA Standard Value **monthly gross total returns** on contemporaneous RF, Mkt-RF, HML and SMB, including an intercept, over **343 months (January 1995–July 2023)**. This is one index time series, so it is a **time-series OLS regression**, notwithstanding the assignment's "pooled panel" wording. Returns and the intercept are in monthly percentage points. Reported t-statistics are conventional OLS, as requested (no Newey–West correction).

Factor	Loading	t-statistic	Economic interpretation
Intercept (monthly %)	-0.0713	-0.91	No statistically detectable positive residual return
RF	0.8646	2.79	Cash-rate sensitivity estimated rather than fixed at one
Market excess return	0.9480	76.99	Near-market equity exposure, not market neutrality

Factor	Loading	t-statistic	Economic interpretation
HML (value)	**+0.3216**	**19.68**	Clear, statistically strong value tilt
SMB (small minus big)	**-0.1744**	**-9.90**	Large-cap tilt consistent with the Standard universe

Exhibit 5B. Factor fingerprint of MSCI USA Standard Value; OLS R2 = **94.84%** (n = 343). The 95% confidence interval for HML is **[0.289, 0.354]**, and for SMB **[-0.209, -0.140]**. Source: supplied Fama–French factors and MSCI gross USD index levels.

Translate the coefficients into a stress test. Holding other regressors fixed, a **+1 percentage-point HML month** predicts approximately **+0.322 percentage points** of MSCI Value return; a **+1 percentage-point SMB month** predicts **-0.174 percentage points**. Thus, an episode in which *small caps outperform large caps* is a headwind for this nominally "value" index even if the pure value spread is unchanged. That is a hidden implementation choice an asset owner should understand: **the index delivers value exposure bundled with a large/mid-cap bias**.

Verdict: as advertised, but not a pure HML trade. The positive HML and negative SMB coefficients match the index's style and universe; the near-unit market loading confirms it remains an equity portfolio. The 94.84% fit means standard factors explain most *in-sample return variation*, not that 94.84% of performance is attributable to stock-picking skill. The remaining variation could reflect sector weights, reconstitution, valuation signals beyond book/price and measurement differences. It is not automatically alpha; the intercept in 5C tests that narrower claim.

5C. Enhancing factor performance

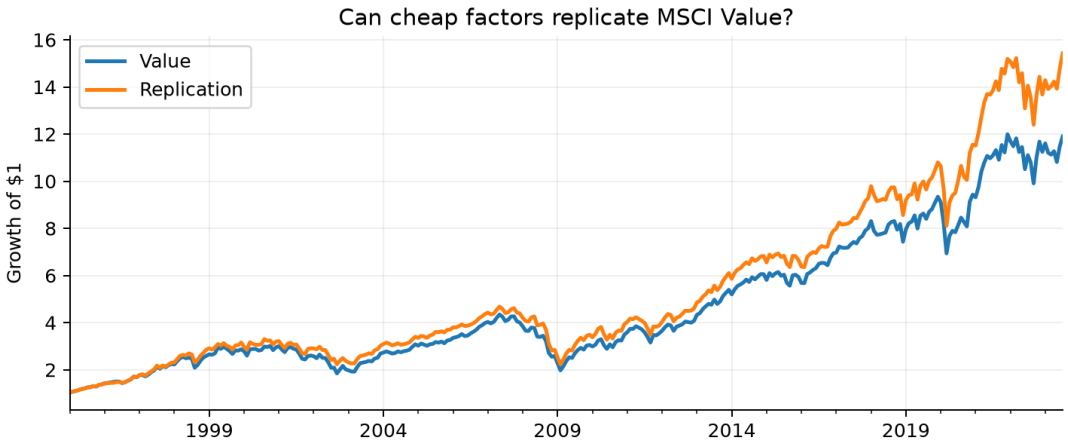
The estimated intercept is **-0.0713% per month** (t = -0.91), approximately **-0.86% per year** under simple annualization. It is negative and statistically indistinguishable from zero (p = 0.366). Thus the data provide **no evidence that MSCI's more elaborate value definition improved risk-adjusted performance** over the fitted generic FF factor combination. This is regression alpha, not a causal test of index-design superiority; the sample includes historical methodology changes and is gross of costs.

5D. Fund replication

The zero-intercept replicating return is 0.8646 RF + 0.9480(Mkt-RF) + 0.3216 HML -0.1744 SMB. The in-sample correlation with MSCI Value is **0.974**; replication is strong in explaining fluctuations, but is not a feasible self-financing investment by itself without specifying factor instruments, financing and collateral.

Metric	MSCI Value	FF replication
Annualized arithmetic return	9.87%	10.73%
Annualized compounded return (CAGR)	9.05%	10.05%
Annualized volatility	15.27%	14.87%
Annualized Sharpe ratio	0.505	0.576

Exhibit 5D. MSCI Value versus in-sample FF replication, Jan 1995–Jul 2023, gross monthly USD returns. Sharpe = sqrt(12) x mean(monthly return - RF) / std(monthly total return); annualized volatility = sqrt(12) x monthly sample standard deviation. The FF replication has higher gross return and Sharpe in-sample, but the regression weights were fitted on the same evaluation period. Before implementation, assess (i) factor-leg trading, shorting and financing costs, (ii) turnover, liquidity and rebalancing frictions, and (iii) availability of suitable investable factor products and their tracking error. A rolling out-of-sample test is needed before selecting the replication in practice.



5E. Relative performance analysis

Each component's monthly active return equals its MSCI gross total return minus MSCI USA Standard's same-month return. The multi-factor index is the equal-weighted monthly average of Value, Minimum Volatility, Equal Weighted and Momentum. Its annualized active return is 12 x mean(active monthly return), and its annualized tracking error is sqrt(12) x sample standard deviation of monthly active return.

Index	Annualized active return	Annualized tracking error	Information ratio
Value	-1.34%	5.15%	-0.261
MinVol	-0.80%	6.38%	-0.125
EqualWeight	+0.59%	5.40%	0.109
Momentum	+2.16%	7.73%	0.279
MultiFactor	+0.15%	3.39%	0.045

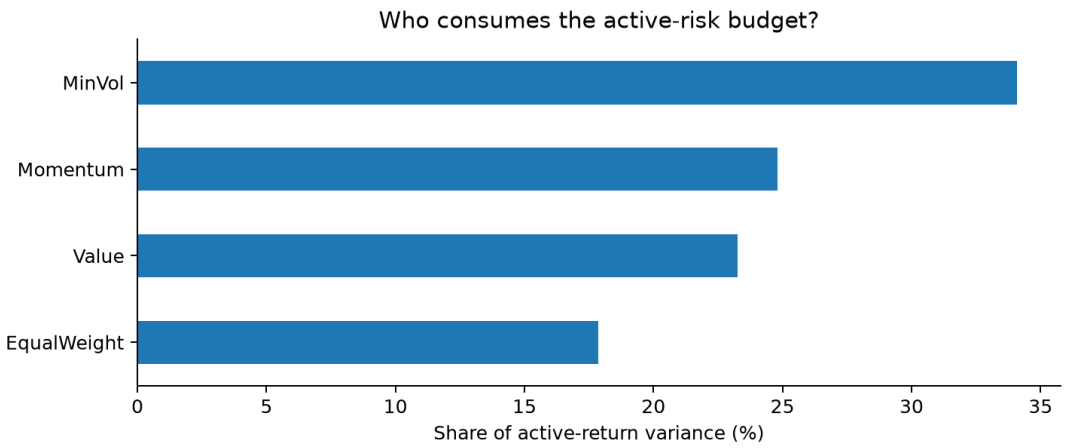
Exhibit 5E. Benchmark-relative performance, Jan 1995–Jul 2023. The multi-factor blend's annualized excess return is only **0.15%** for **3.39%** tracking error, producing an information ratio of **0.045**. This is economically weak active efficiency: the blend improves some absolute-risk characteristics relative to individual factors, but adds little return per unit of benchmark-relative risk.

5F. Risk decomposition

With active factor returns ai and equally weighted active blend ap = 1/4sum iai, its variance satisfies Var(ap) = 1/4sum iCov(ap,ai). We attribute each component's share of the blend's active-return variance as Cov(ap,ai)/[4 Var(ap)]. These signed contributions sum to 100% and reflect both standalone volatility and cross-factor correlations.

Factor	Share of active-return variance
Value	23.26%
MinVol	34.08%
EqualWeight	17.86%
Momentum	24.81%

Exhibit 5F. Covariance-based tracking-error variance attribution, Jan 1995–Jul 2023. **Minimum volatility (34.1%)** is the largest contributor, followed by Momentum (24.8%). Although minimum volatility is attractive as a stand-alone defensive allocation, it can create substantial benchmark-relative risk because its defensive tilt differs from the cap-weighted equity benchmark. Under limits to arbitrage, low-beta stocks can offer attractive risk-adjusted standalone returns, but that does not imply low tracking error for a benchmark-constrained investor. Thus the appropriate choice depends on the investor's mandate: standalone for absolute-risk control, or a more carefully sized tilt for benchmark-relative mandates.



Sources and notes. Supplied workbook `class-project-data-ff-msci-parts-1-and-5.xlsx`: FF monthly RF, Mkt-RF, SMB, HML and MSCI gross USD index levels; MSCI methodology: https://www.msci.com/eqb/methodology/meth_docs/MSCI_GIMIVGMethod_Feb2021.pdf. MSCI monthly returns are computed from adjacent month-end index levels. The factor replicating return omits the fitted intercept by design. All results gross of fees/costs, using in-sample OLS. The arithmetic return is mean monthly return x 12; CAGR is compounded monthly growth annualized. The supplied MSCI series and FF factor data end in July 2023.